

OKTYABR'SKIY GORODOK, Russian Federation

WMO: 341630

Lat: 51.633N

Lon: 45.450E

Elev: 201

StdP: 98.93

Time Zone: 4.00 (E04)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -26.4	(c) -23.5	(d) -29.4	(e) 0.3	(f) -26.3	(g) -26.2	(h) 0.4	(i) -23.4	(j) 9.1	(k) -8.1	(l) 8.3	(m) -9.0	(n) 1.9	(o) 80	(p) 0.564

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 11.9	(c) 34.4	(d) 20.0	(e) 32.3	(f) 19.2	(g) 30.4	(h) 18.7	(i) 20.9	(j) 31.0	(k) 20.2	(l) 29.9	(m) 19.5	(n) 28.5	(o) 3.3	(p) 100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.9	13.1	23.4	16.9	12.4	23.1	16.1	11.8	23.0	61.2	30.9	58.6	30.1	56.3	28.5	23.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 8.5	(b) 7.5	(c) 6.8	(e) -30.6	(f) 36.4	(g) 2.3	(h) 1.9	(i) -32.2	(j) 37.8	(k) -33.6	(l) 38.8	(m) -34.9	(n) 39.9	(o) -36.5	(p) 41.2		
(a) 8.5	(b) 7.5	(c) 6.8	(e) -30.7	(f) 22.3	(g) 2.3	(h) 1.1	(i) -32.3	(j) 23.1	(k) -33.7	(l) 23.8	(m) -34.9	(n) 24.4	(o) -36.6	(p) 25.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	6.4	-11.0	-9.7	-3.8	7.4	15.9	20.2	22.3	20.8	14.1	5.6	-0.6	-5.8
	DBStd	12.85	7.04	6.92	5.44	5.06	4.00	4.36	3.85	4.40	4.06	4.86	5.16	5.86
	HDD10.0	2716	651	552	427	112	2	1	0	0	9	154	318	488
	HDD18.3	4718	909	786	686	329	99	29	9	23	139	396	568	747
	CDD10.0	1389	0	0	0	32	184	306	381	336	131	17	1	0
	CDD18.3	349	0	0	0	0	22	84	132	101	11	0	0	0
	CDH23.3	3897	0	0	0	8	242	896	1473	1162	117	0	0	0
	CDH26.7	1589	0	0	0	0	54	349	656	504	27	0	0	0
Wind	WSAvg	3.2	3.1	3.6	3.4	3.4	3.1	3.0	2.8	2.7	3.0	3.2	3.2	3.6
Precipitation	PrecAvg	428	33	22	20	26	43	44	52	49	37	34	37	30
	PrecMax	692	72	73	51	60	105	110	108	242	96	91	75	71
	PrecMin	226	3	1	0	3	8	10	3	0	2	0	10	8
	PrecStd	108	17	15	11	16	30	28	27	45	25	22	20	14
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.4	2.6	11.2	24.5	30.3	35.4	36.9	37.0	29.6	20.1	12.9	6.1
		MCWB	0.7	1.5	7.0	14.4	17.0	20.5	21.2	20.1	16.9	12.3	10.7	5.6
	2%	DB	0.6	1.6	6.3	21.8	28.1	32.5	34.6	34.0	26.5	17.6	10.4	3.2
		MCWB	0.1	0.7	4.1	12.8	16.0	19.5	19.9	19.7	15.7	11.6	8.8	2.5
	5%	DB	0.0	0.7	4.1	19.2	26.2	30.6	32.6	31.7	24.0	15.1	8.4	1.9
		MCWB	-0.3	0.0	2.3	11.5	15.6	19.2	19.7	18.7	14.5	10.6	7.2	1.3
	10%	DB	-1.1	-0.4	2.5	16.5	23.8	28.5	30.4	29.5	21.6	12.7	6.2	0.9
		MCWB	-1.6	-0.9	1.1	10.0	14.6	18.5	18.8	18.1	13.6	9.3	5.3	0.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.0	1.9	7.2	15.5	18.6	22.3	21.9	21.6	18.4	14.2	10.7	5.7
		MCDB	1.3	2.3	10.3	23.1	26.4	31.2	34.2	32.8	27.2	16.2	12.1	6.0
	2%	WB	0.2	0.8	4.6	13.7	17.4	21.0	20.7	20.4	16.5	12.5	8.9	2.6
		MCDB	0.5	1.4	6.4	20.5	25.5	30.0	31.3	31.2	23.7	16.2	10.2	3.0
	5%	WB	-0.3	0.0	2.5	12.1	16.2	20.1	20.1	19.6	15.2	11.0	7.3	1.3
		MCDB	0.0	0.6	3.7	17.8	24.2	28.5	30.3	29.7	21.2	14.9	8.4	1.8
	10%	WB	-1.5	-0.9	1.3	10.4	15.2	19.2	19.5	18.7	14.4	9.7	5.3	0.4
		MCDB	-1.1	-0.3	2.3	16.0	22.5	26.5	28.8	27.8	19.9	12.5	6.1	0.8
Mean Daily Temperature Range	5% DB	MDBR	6.2	6.3	7.3	10.1	10.8	11.3	11.9	12.1	9.9	7.6	5.0	4.7
		MCDBR	5.2	4.4	7.6	14.6	14.0	13.5	14.5	14.5	14.1	10.9	6.0	3.7
		MCWBR	5.2	4.2	5.5	7.5	6.1	5.2	5.2	5.3	6.5	6.2	4.9	3.6
	5% WB	MCDBR	5.1	4.5	6.8	14.0	12.0	12.3	12.8	13.0	11.8	9.8	5.8	3.7
		MCWBR	5.2	4.4	5.1	7.8	5.9	5.2	5.3	5.3	5.9	6.8	5.0	3.6
Clear-Sky Solar Irradiance	taub	0.298	0.331	0.364	0.430	0.400	0.388	0.404	0.423	0.376	0.351	0.327	0.300	
	taud	2.107	2.025	2.059	2.094	2.297	2.366	2.319	2.250	2.358	2.365	2.275	2.136	
	Ebn at Noon	677	756	812	802	854	867	844	803	804	749	651	604	
	Edh at Noon	77	111	131	144	123	116	120	122	98	80	66	64	
All-Sky Solar Radiation	RadAvg	0.93	2.03	3.38	4.34	5.71	6.14	6.09	5.15	3.51	1.82	0.79	0.55	
	RadStd	0.21	0.33	0.57	0.39	0.54	0.54	0.50	0.39	0.44	0.27	0.14	0.15	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)	+0.69	N/A	N/A	+1.25	N/A	N/A	N/A	N/A	+179	+77

Nomenclature: See separate page